

Arunabha Majumder

Mechanical Engineer — Robotic Actuators, Joint Kinematics, Prototyping & Test Rigs

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LinkedIn: [linkedin.com/in/arunabha-majumder-681264107](https://www.linkedin.com/in/arunabha-majumder-681264107) • Portfolio: arunabha22.github.io/arunabha-majumder

October 7, 2026

Hiring Team
Cell Labs
Berlin, Germany

Application: Mechanical Engineer – Humanoid Systems

Dear Cell Labs Team,

I am applying for the Mechanical Engineer position. I am a mechanical and mechatronics engineer finishing my PhD at Aalborg University, and what I enjoy most is exactly what you describe: designing robotic joints and actuators, getting a first prototype built quickly, and improving it on the bench until it works.

In my PhD, I designed and built a robotic shoulder joint for an exoskeleton. Instead of a large motor carrying the full gravity torque, I combined a parallel spring with a small 6 Nm motor, sized so the torque and load requirements were met across the motion. I designed the components and assemblies in SolidWorks and Fusion 360, used FEA for stress analysis and weight optimisation, made the parts through 3D printing and machining, and integrated the motor, encoder, force sensors, DAQ and embedded control. The design cut energy consumption by more than 25% and won the Best Paper Award at IFToMM ISRM 2026.

I understand how kinematics and actuators come together in a working robot. On a 5-bar planar parallel robot, I modelled the kinematics and estimated the end-effector force from variable-stiffness actuators through the robot's Jacobian, validated against a force sensor to below 10% RMS error. I build my own test rigs, track down mechanical, electrical and control problems on the bench, and I have built machines since my student years, including a Go-Kart from the ground up.

My hardware so far has been prototypes rather than products in series production, and I have not yet worked with injection moulding, sheet metal or supplier ramp-up, or used Onshape. These are exactly the skills I want to build, and a team that takes humanoid robots from CAD to production under one roof is the right place to do it. I am open to relocating to Berlin.

I would welcome the opportunity to show you my work in a technical deep-dive. My portfolio is linked above, and references are available upon request.

Kind regards,
Arunabha Majumder